

Academic Course Finder

Final Project

Team:

- ***Brayan Alejandro Riveros Rodriguez - 20201020084***
- ***Carlos Andrés Pescador Castro - 20182020139***
- ***Cristian Santiago Ríos Vásquez - 20201020107***



INTRODUCTION

Students and teachers often have trouble finding course details or knowing who teaches each class.

This happens because academic data is spread across many systems and formats.

As a result, it becomes slow and difficult to get the right information, which affects academic planning.



PRIMARY OBJECTIVE

Deliver a web-based system for
course finding

VALUE PROPOSITIONS

- Search & Browse fast for course and professor information
- Creating link and unlink course-professor relationship.
- Admin User Interface for data management



ACADEMIC COURSE FINDER

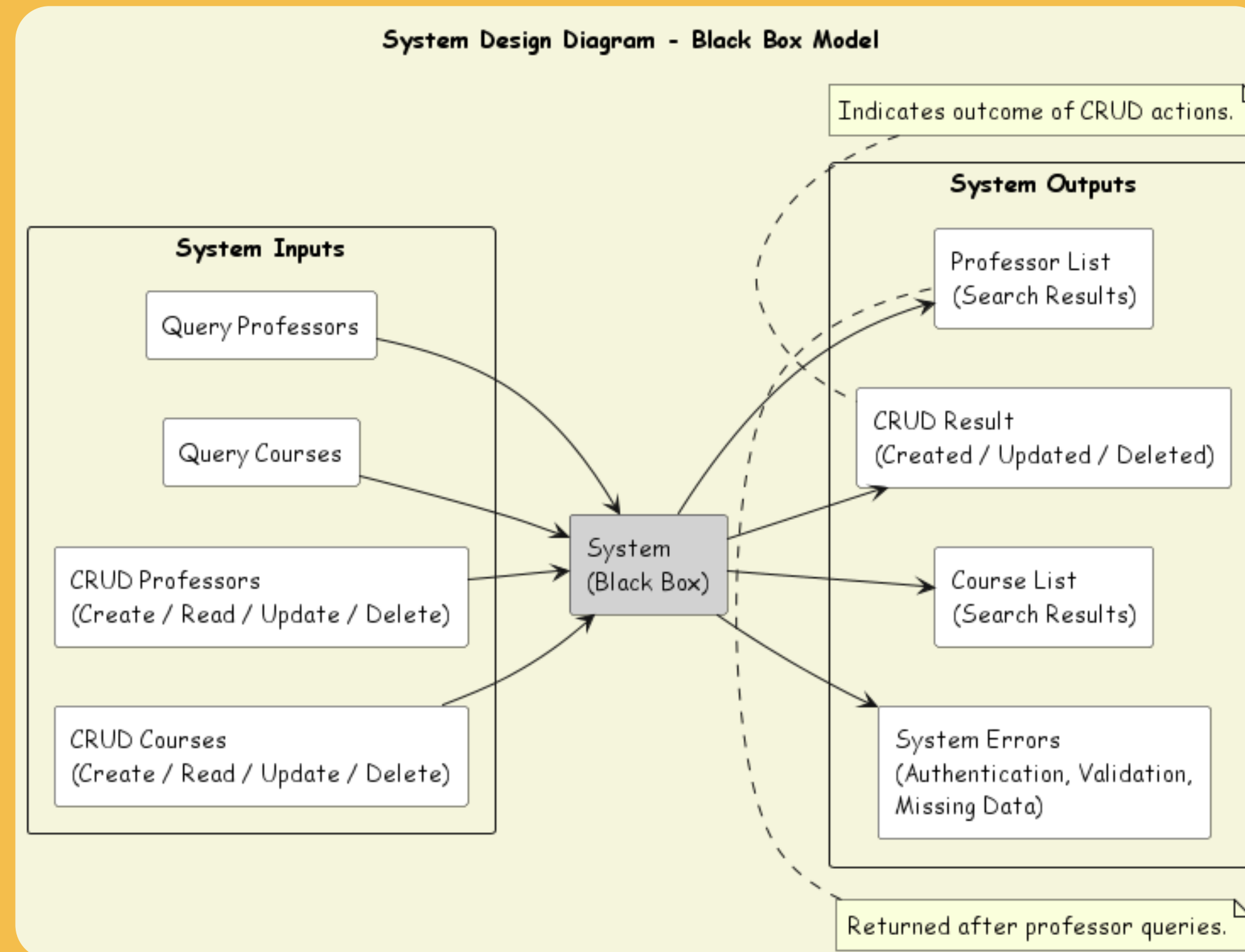
REQUIREMENTS AND USER STORIES IDENTIFIED

- Registration of new teachers and their relationships
- Registration of new subjects
- Home page and access flow for administrator and user
- Link professor to one or multiple courses
- Search courses by professor and professors by Course



SYSTEM VISION

A system that centralizes professor and course information to enable queries and administrator control



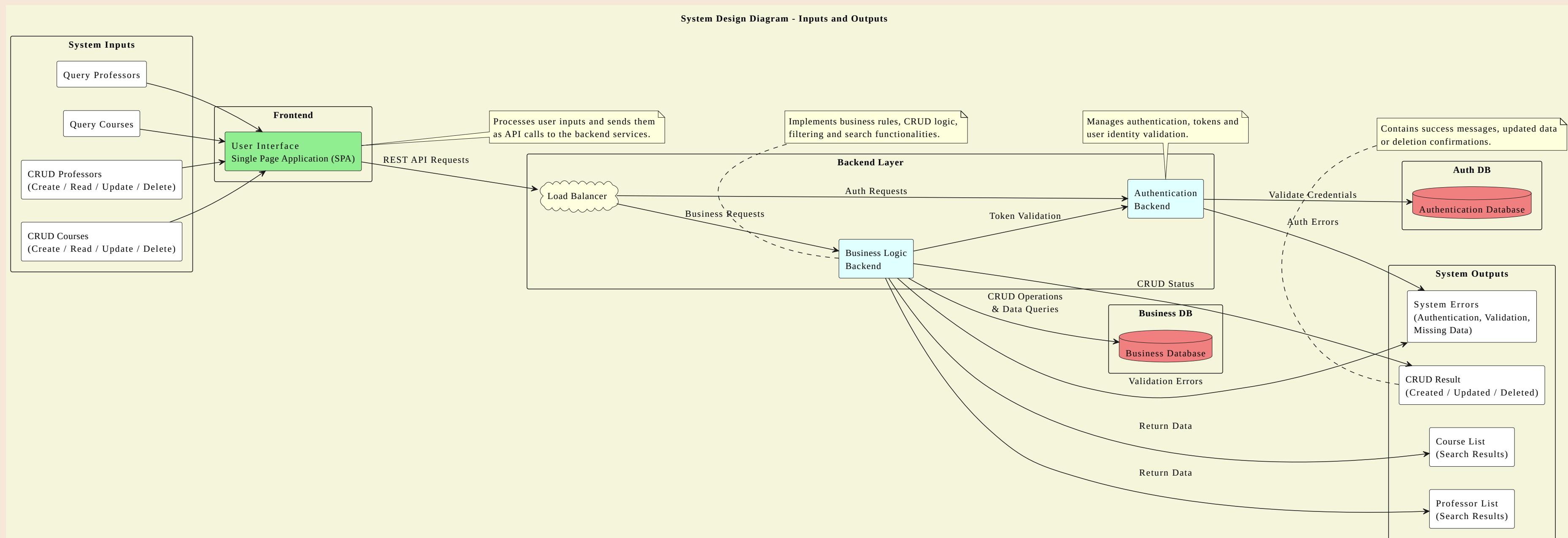
PROBLEM

Access to academic information is difficult and not centralized.

PROPOSED SOLUTION

The Academic Course Finder Web Application solves this by creating one platform where:

- Admins manage teachers/courses
- Users search courses without login



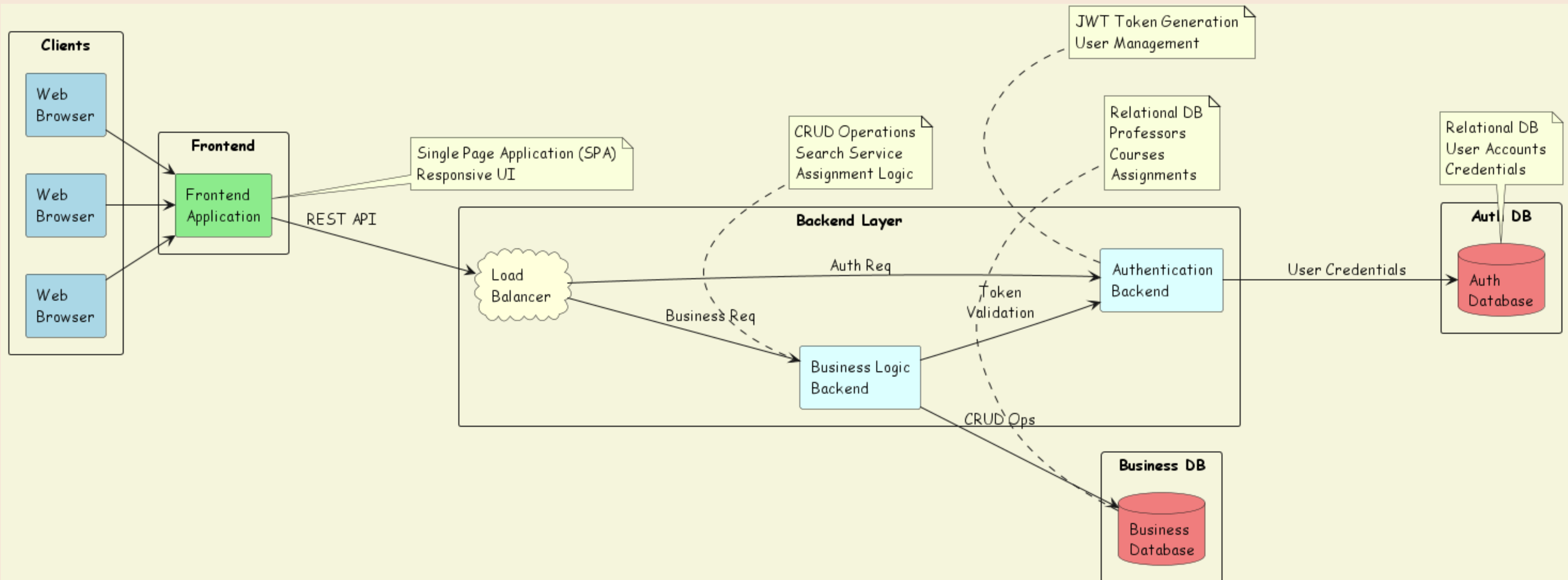
PROBLEM

Access to academic information is difficult and not centralized.

PROPOSED SOLUTION

The Academic Course Finder Web Application solves this by creating one platform where:

- Admins manage teachers/courses
- Users search courses without login



DEMO TIME!



Watch video on YouTube

Error 153

Video player configuration error



[Demo en Youtube](#)

RESULTS

SUMMARY OF THE COMPLETED SYSTEM DELIVERABLES

- Fully implemented web application for course and professor management
- Functional search and browsing experience across all academic data
- Completed authentication flow with role-based access for admin users
- Finalized data model and integrated relational databases
- End-to-end API layer implemented and validated
- Frontend UI finalized with consistent interaction patterns
- System documentation delivered: architecture, API specs, and user guides

UNIT TESTS

Core components validated through automated unit testing

TEST RESULTS

- Business logic functions tested with consistent and correct outputs
- Validation rules verified for all professor and course operations
- Error handling and edge cases confirmed across key components
- High test reliability with stable and repeatable results

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.udistrital.authback.service.AuthServiceTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation:
https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (/home/runner/.m2/repository/net/bytebuddy/byte-buddy-agent/1.17.8/byte-buddy-agent-1.17.8.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnabledDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.699 s -- in com.udistrital.authback.service.AuthServiceTest
[INFO] Running com.udistrital.authback.security.JwtUtilTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.100 s -- in com.udistrital.authback.security.JwtUtilTest
[INFO] Running com.udistrital.authback.AuthBackApplicationTests
Warning: Tests run: 1, Failures: 0, Errors: 0, Skipped: 1, Time elapsed: 0 s -- in com.udistrital.authback.AuthBackApplicationTests
[INFO]
[INFO] Results:
[INFO]
Warning: Tests run: 5, Failures: 0, Errors: 0, Skipped: 1
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 23.293 s
[INFO] Finished at: 2025-11-20T09:11:40Z
```

```
===== test session starts =====
platform linux -- Python 3.11.14, pytest-7.4.4, pluggy-1.6.0
rootdir: /home/runner/work/Software-engineering-seminar/Software-engineering-seminar/Workshop-4/python-backend
configfile: pyproject.toml
plugins: anyio-3.7.1, cov-7.0.0, asyncio-0.21.2
asyncio: mode=Mode.AUTO
collected 26 items

tests/test_course_service.py ..... [ 50%]
tests/test_professor_service.py ..... [100%]

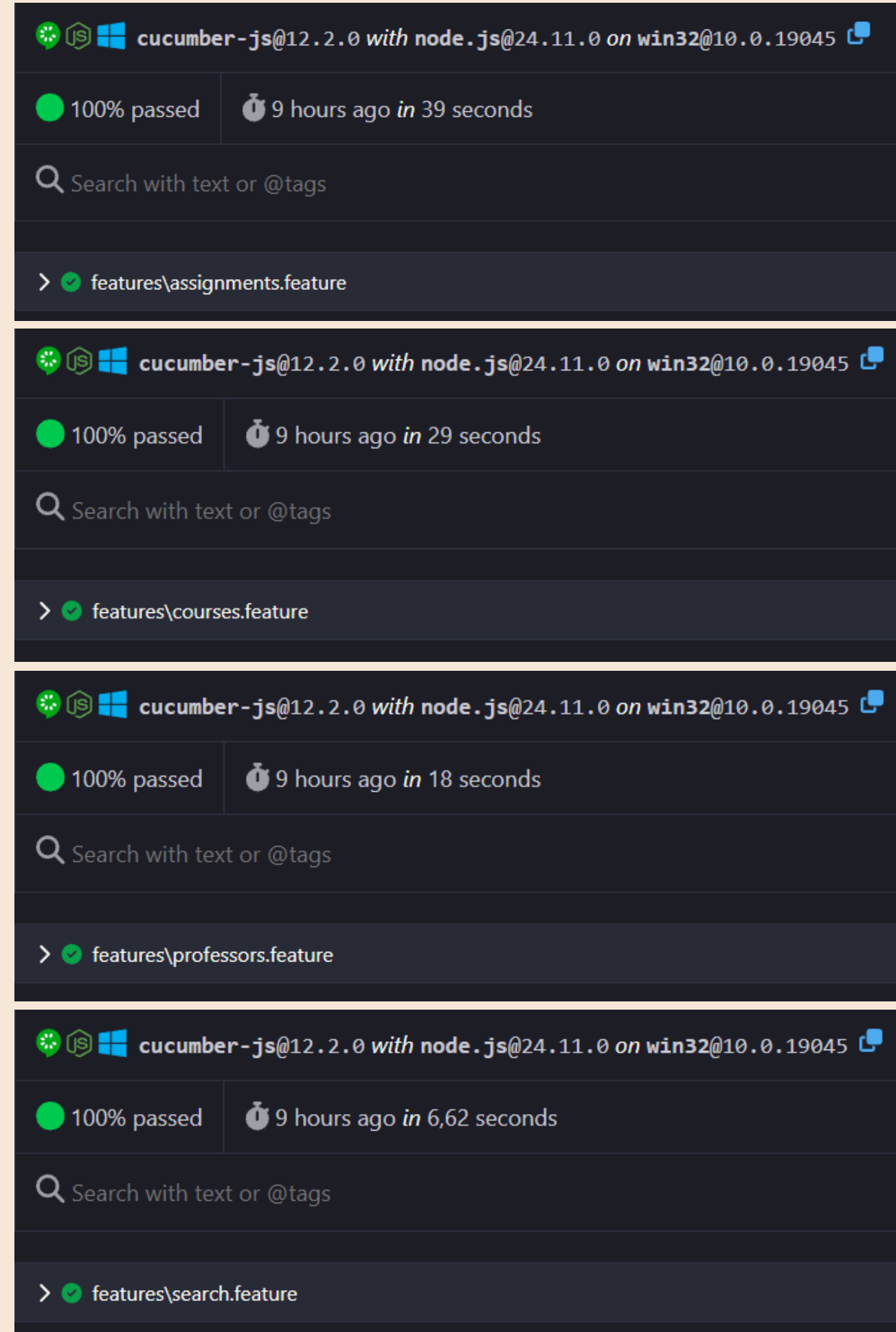
===== 26 passed in 0.45s =====
```

ACCEPTANCE TESTS

System delivered according to the defined acceptance criteria

TEST RESULTS

- All core features validated: search, browse, and management of professors and courses
- UI tested for correct behavior across main views
- API responses confirmed for accuracy, structure, and expected data
- End-to-end scenarios executed with consistent and reliable outputs

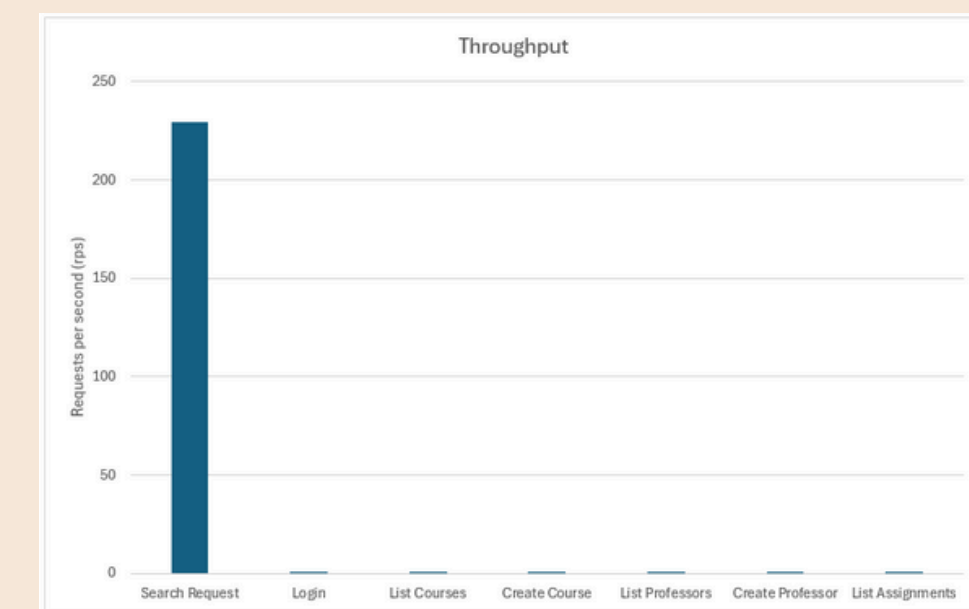
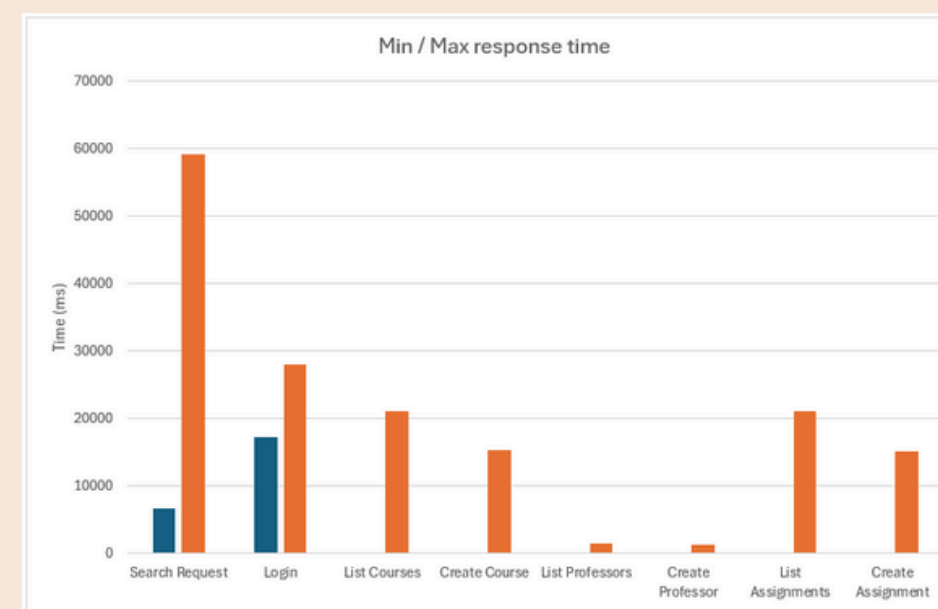
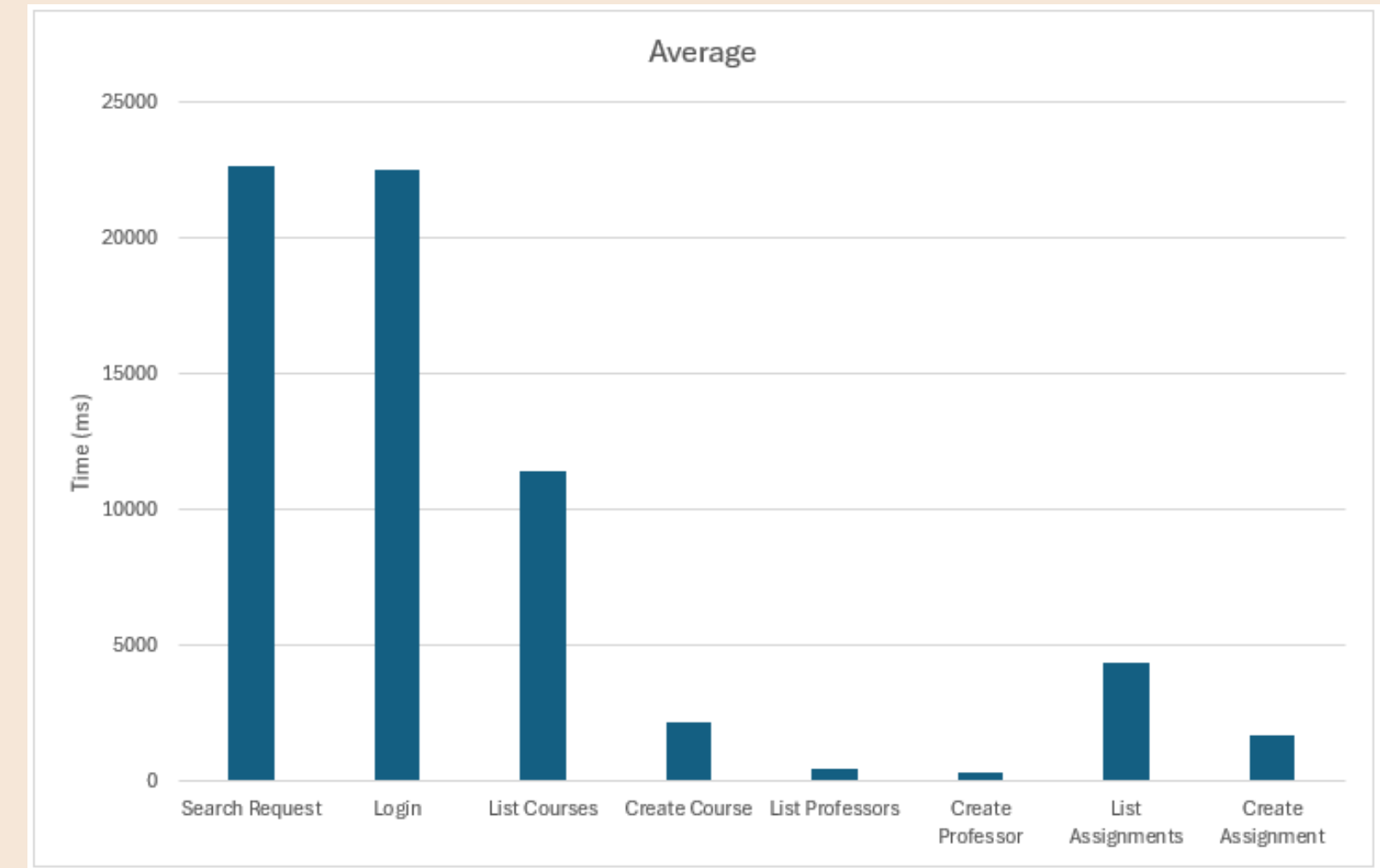


API STRESS TESTS

System behavior validated under high load and peak usage conditions

TEST RESULTS

- Stable performance under high volumes of search and CRUD operations
- Backend services-maintained response consistency during peak request bursts
- Database sustained concurrent read/write operations without degradation
- No critical failures detected under maximum simulated load



REFLECTION AND EVALUATIONS

UPCOMING DELIVERABLES

- All user stories were implemented and tested, with a effective scrum methodology for our team coordination
- The usage of taiga platform tool for methodology, offered us a clear visibility and traceability
- Performance bottlenecks were identified, in future development needs scaling strategies

THANK YOU!